


Lecture 11 Classification Theorem

Can we list all finite groups?

isomorphic
 $\cong = \cong$

Classification Thm for ^{finite} cyclic groups

If G is a cyclic group of order n then

$$G \cong \mathbb{Z}_n.$$

Classification for finitely generated abelian groups

Classification for finite abelian groups

If G is a finite abelian group then $G \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \dots \times \mathbb{Z}_{n_k}$
where n_i is a power of a prime.

Ex 1

let G be an abelian group with $|G| = 36$
 $= 2^2 \cdot 3^2$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$$

$$\mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$$

$$\mathbb{Z}_4 \times \mathbb{Z}_9$$

} list of all
the abelian groups
of order 36.

$$\mathbb{Z}_6 \times \mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$$

order 2

$$\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$$

$\downarrow$
(x,y)

x order 2 in $\mathbb{Z}_2$
y order 3 in $\mathbb{Z}_3$

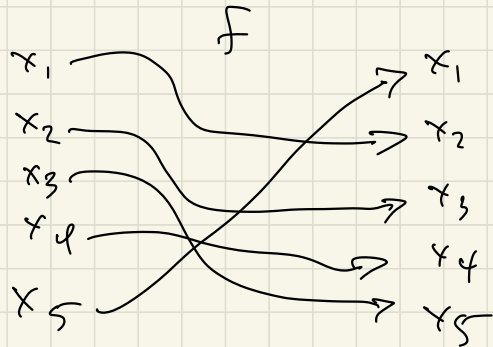
Cayley's Theorem

Let G be a finite group of order n , then

G is isomorphic to a subgroup of S_n , the symmetric group on n elements.

S_X the group of bijections on a set X

Eg)



$$X = \{x_1, \dots, x_5\}$$

$f: X \rightarrow X$ bijection

$$S_X = \{f: f: X \rightarrow X \text{ bijection}\}$$

$(S_X, \text{composition})$ is a group

identity $f(x_i) = x_i$ for all i

inverse ~~Let~~ what I just said

multiplication $f \circ g(x) = f(g(x))$

Is S_n the same as S_X ?

$S_n \cong S_X$ where $|X| = n$

Proof of Cayley's Theorem

want to show that G is isomorphic to a subgroup of S_G .

Let $L_a: G \rightarrow G$ by $L_a(g) = a \cdot g$ where $a \in G$

Claim L_a is a bijection for all $a \in G$

onto Let $x \in G$ $L_a(a^{-1}x) = a(a^{-1}x) = x$ ✓

1 to 1 Suppose $L_a(x) = L_a(y) \Rightarrow ax = ay$
 $\Rightarrow x = y$ (applying a^{-1})
□

In particular $L_a \in S_G$

$\mathcal{L} = \{L_a : a \in G\}$ a set of bijections

$\mathcal{L} \subseteq S_G$ is $\mathcal{L} < S_G$?

Claim $\mathcal{L}$ is a subgroup of S_G under composition

identity L_e $L_e \circ L_a(x) = L_e(ax)$ for all $x \in G$
 $= eax$
 $= ax = L_a(x)$

So $L_e \circ L_a \in \mathcal{L}$

L_e is a left identity

closed under mult

Let $a, b \in G$

$$\begin{aligned} L_a \circ L_b(x) &= L_a(bx) = a(bx) = (ab)x \\ &= L_{ab}(x) \end{aligned}$$

So $\mathcal{L}$ is a subgroup of S_G .

Claim $G \cong \mathcal{L}$

PF Let $L : G \rightarrow \mathcal{L} = \{L_a : a \in G\}$

$$\text{by } L(a) = L_a$$

preserves id $L(e) = L_e$ ← this is identity in $\mathcal{L}$

preserve mult want to show $L(a \cdot b) = L_a \circ L_b$

$$L(a \cdot b) = L_{ab} \quad L_{ab}(x) = abx \quad \text{for all } x$$

$$L_a \circ L_b(x) = L_a(bx) = abx \quad \text{for all } x$$

$$\text{so } L_{ab} = L_a \circ L_b$$

one to one Let $L(a) = L(b)$ then $L_a = L_b$ is $\mathcal{L}$

so $L_a(x) = L_b(x)$ for all $x \in G$

in particular it's true for $x = e$

$$\text{so } L_a(e) = L_b(e)$$

$$\Rightarrow ae = be \Rightarrow a = b \quad \checkmark$$

onto suppose $L_a \in \mathcal{L}$ $L_a = L(a)$ $\checkmark$

$$\text{So } G \cong \mathcal{L} < S_G \quad \square$$

$$S_G \cong S_n$$

where $n = |G|$.

show given $L_g \in \mathcal{L}$ there exists $L_{g^{-1}} \in \mathcal{L}$

claim $L_g^{-1} = L_{g^{-1}}$

$$L_{g^{-1}} \circ L_g(x) = L_{g^{-1}}(gx) = g^{-1}gx = x \\ \text{left inverse} \qquad \qquad \qquad = L_e(x)$$

Cauchy's Theorem Let G be a finite group.

If a prime p divides $|G|$ then G has
an element with order p .